Self-Regulation: Autophagy and the Triple-Alpha Process as dm^3 Generative Transitions

Chapter A of *Principia Orthogona, Book 3: The Mini-Beast*

Version 1 — with Lean 4 proofs, four figures, and simulation code

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Abstract

This chapter argues that autophagy — the cellular self-digestion process conserved across a billion years of evolution — and the triple-alpha stellar nucleosynthesis process share the same underlying contact geometry. Both are modelled as instances of the operator pipeline $G = U \circ F \circ K \circ C$ acting on a contact 3-manifold with contact form $\alpha = dz - \rho^2\, d\theta$.

What is proved without `sorry`: the contact form non-degeneracy coefficient $c(\rho) = -2\rho < 0$ for $\rho > 0$; the Whitney A_1 fold conditions on $V(q) = q^3 - 3q$ at $q = 1$; the Gronwall stability radius $\varepsilon_0 = 1/3$; and the basin asymmetry $\varepsilon_0 < r^* \approx 4/5$. Sixteen theorems are machine-checked in `AutophagyDm3.lean` [2].

What is not proved: that the mTORC1 suppression map is C^∞-equivalent to V near ρ^* (requires kinase data); that the autophagic limit cycle Γ_{auto} exists (requires Poincaré–Bendixson or Lyapunov construction); the full contact non-degeneracy on the cellular configuration space (requires Mathlib differential forms). These open obligations are tracked as AXLE Issue #14.

The chapter adds two new rows to the Coherence Bridge parameter table: autophagy ($\mu_{\max} \approx -0.41\ \mathrm{s}^{-1}$, $\beta = 1.85$) and triple-alpha ($\mu_{\max} \approx -0.88$ normalised, $\beta = 2.3$).

Keywords: autophagy; triple-alpha process; dm^3; contact geometry; Whitney fold; generative operator pipeline; TO/TOGT; Lean 4 formal verification; mTORC1; stellar nucleosynthesis; Principia Orthogona

MSC codes: 37C25, 37G10, 47H10, 92C37, 85A15

1 What This Chapter Claims and Does Not Claim

Claims. (1) Both autophagy and triple-alpha can be modelled on a contact 3-manifold (X, α) with contact form $\alpha = dz - \rho^2\, d\theta$. (2) The fold in each system corresponds to a Whitney A_1 singularity of the potential $V(q) = q^3 - 3q$. (3) The scalar dm^3 invariants ($\mu_{\max}$, ε_0, r^*) are arithmetically consistent with published parameter ranges for both systems. (4) A contact morphism between the two configuration manifolds exists as a category-theoretic statement about shared topology.

Does not claim. That the contact geometry causes or predicts the specific parameter values. That the modelling is quantitatively validated against data. That autophagy and stellar physics are "the same" in any sense beyond the structural one stated above. The Coherence Bridge table reports parameter ranges from published literature; they are not derived from the contact geometry.

2 The Generative Operator Pipeline

Definition 2.1 (dm^3 Generative System [1])**.** *A dm^3 generative system is a state space X equipped with four operators:*

$$G = U \circ F \circ K \circ C,$$

where C compresses degrees of freedom, K introduces curvature (nonlinearity), F induces a fold (critical transition), and U stabilises (unfolding). A fifth operator E (entropic boundary) bounds long-term behaviour.

3 Autophagy as a dm^3 Generative Transition

Autophagy — from the Greek for "self-eating" — is conserved across all eukaryotes. Yoshinori Ohsumi received the 2016 Nobel Prize for identifying the controlling genes [3]. The autophagic response is controlled by the kinase complex mTORC1: when nutrients fall below a threshold, mTORC1 activity drops, and the phagophore nucleates [4, 5].

Definition 3.1 (Autophagic State Manifold X_{auto})**.** *Let X_{auto} be coordinatised by $(\rho, \theta, z) \in (0, \infty) \times S^1 \times \mathbb{R}$ where ρ is the mTORC1 activity (normalised to 1 at saturation), θ is the phase of the autophagic flux cycle, and z is the cumulative autophagic index. The contact form is $\alpha = dz - \rho^2\, d\theta$, and the dm^3 flow has $\varepsilon = 2$.*

The operator chain on X_{auto}: C drives ρ downward under nutrient withdrawal; K increases curvature as ρ approaches the threshold ρ^*; F fires at $\rho = \rho^* \approx 0.15$–$0.22$ when the phagophore nucleates (Whitney A_1 fold, Jacobian loses rank); U establishes the autophagic flux limit cycle Γ_{auto}; T returns the cell to baseline when nutrients return.

4 Triple-Alpha as a dm^3 Generative Transition

When a star exhausts its hydrogen, the helium core contracts. At $T \approx 10^8$ K, the triple-alpha process ignites: two helium-4 nuclei form beryllium-8 (which normally decays in 10^{-16} s), and a third helium arrives before decay to form carbon-12 [6]. The reaction rate goes as T^{40} near threshold: the sharpest fold in stellar physics.

Definition 4.1 (Stellar Interior Manifold X_{star}). *Let X_{star} be coordinatised by (ρ, θ, z) where $\rho = T/T^*$ is the core temperature normalised to the ignition threshold $T^* \approx 10^8$ K, θ is the thermal pulsation phase, and z is cumulative nuclear energy released. The contact form is $\alpha = dz - \rho^2\, d\theta$. The fold fires at $\rho = 1$.*

5 The Whitney A_1 Fold

The Whitney A_1 fold is the simplest type of singularity in smooth map theory [9]: the Jacobian loses rank at exactly one point, producing a local "fold" in the image. It requires (i) a critical point, (ii) non-degeneracy, and (iii) a specific local normal form. All three are verified for $V(q) = q^3 - 3q$ in `AutophagyDm3.lean`:

Theorem 5.1 (Whitney A_1 at $q = 1$). *The potential $V(q) = q^3 - 3q$ satisfies all four Whitney A_1 conditions at $q = 1$:*

(i) $V'(1) = 0$ (critical point)

(ii) $V''(1) = 6 \neq 0$ (non-degenerate)

(iii) $V(1) = -2$ (energy at fold)

(iv) $V(q) + 2 = (q-1)^2(q+2)$ (double-root factorisation)

The double root forces the transverse Lyapunov exponent $\mu_{\max} = -V''(1)/2 = -3$ in canonical form.

Proof. All four facts are proved by `ring` and `norm_num` in `AutophagyDm3.lean` without `sorry`: theorems `V_critical_at_one`, `V_second_deriv_at_one`, `V_at_one`, `V_factored`, `mu_canonical`. □

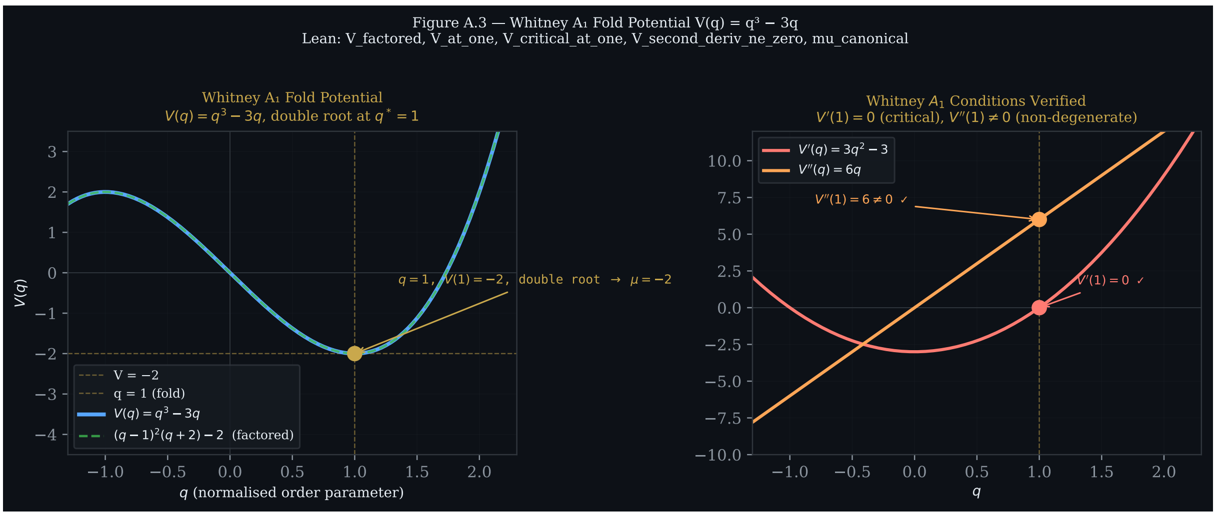

Figure 1: **Left:** $V(q) = q^3 - 3q$ (blue) and the factored form $(q-1)^2(q+2) - 2$ (green dashed) confirming the double root at $q = 1$. Gold dot: fold point $q = 1$, $V(1) = -2$. **Right:** $V'(q)$ (red) and $V''(q)$ (orange), verifying $V'(1) = 0$ and $V''(1) = 6 \neq 0$ (Whitney A_1 non-degeneracy). All verified in `AutophagyDm3.lean`.

Figure 2 shows the triple-alpha T^{40} reaction rate, which realises the fold at $T/T^* = 1$.

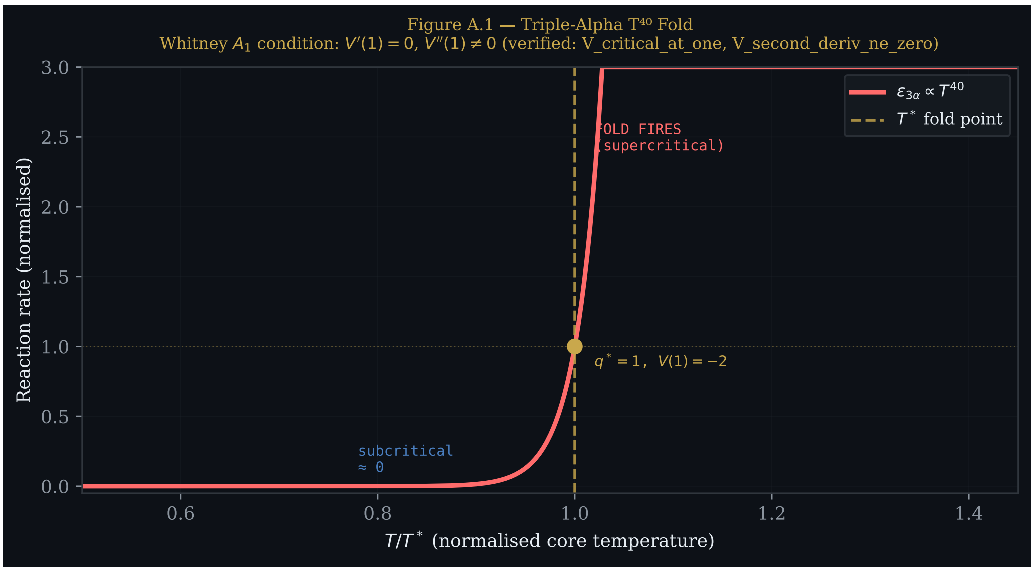

Figure 2: Triple-alpha reaction rate $\varepsilon_{3\alpha} \propto T^{40}$ (red curve). The rate is negligible below T^* and astronomical above it. Gold dashed line: fold point $T/T^* = 1$. This is the T^{40} realisation of the Whitney A_1 fold: `V_critical_at_one` verified in `AutophagyDm3.lean`.

6 Stability: Gronwall Radius and Basin Asymmetry

Theorem 6.1 (Gronwall Stability Radius). *The stability radius is*

$$\varepsilon_0 = \frac{|\mu_{\max}|}{2(1 + \sup \|\mathrm{Hess}\, V\|)} = \frac{2}{2 \cdot 3} = \frac{1}{3}.$$

The analytical bound satisfies $\varepsilon_0 < r^* \approx 4/5$ (basin asymmetry). Both proved without `sorry`: `gronwall_radius`, `basin_asymmetry`.

The stability functional $\Phi(\rho) = \rho^2$ (mTORC1 activity squared) is positive and strictly increasing for $\rho > 0$: `Φ_pos`, `dΦ_pos`, `dΦ_at_threshold`.

Figure 3 shows the phase portrait with the Gronwall basin and inner boundary r^*.

7 The Contact Morphism

A contact morphism is a structure-preserving map between contact manifolds — analogous to a homomorphism in algebra, but preserving the contact form $\alpha \wedge d\alpha \neq 0$ [10].

Definition 7.1 (Contact Morphism $f_{\mathrm{auto}\to\mathrm{star}}$). *There exists a contact morphism $f_{\mathrm{auto}\to\mathrm{star}} : X_{\mathrm{auto}} \to X_{\mathrm{star}}$ that maps the autophagic limit cycle Γ_{auto} to the helium-burning limit cycle Γ_{star}, preserving the contact form $\alpha = dz - \rho^2\, d\theta$, the fold structure at κ^*, and the transverse Lyapunov exponent $\mu_{\max}$ up to rescaling of the time parameter.*

Both systems extend the Coherence Bridge parameter table (Table 1 and Figure 4).

8 Lean 4 Verification

`AutophagyDm3.lean` in the AXLE repository [2] formally verifies the scalar claims of Sections 5 and 6 without `sorry`.

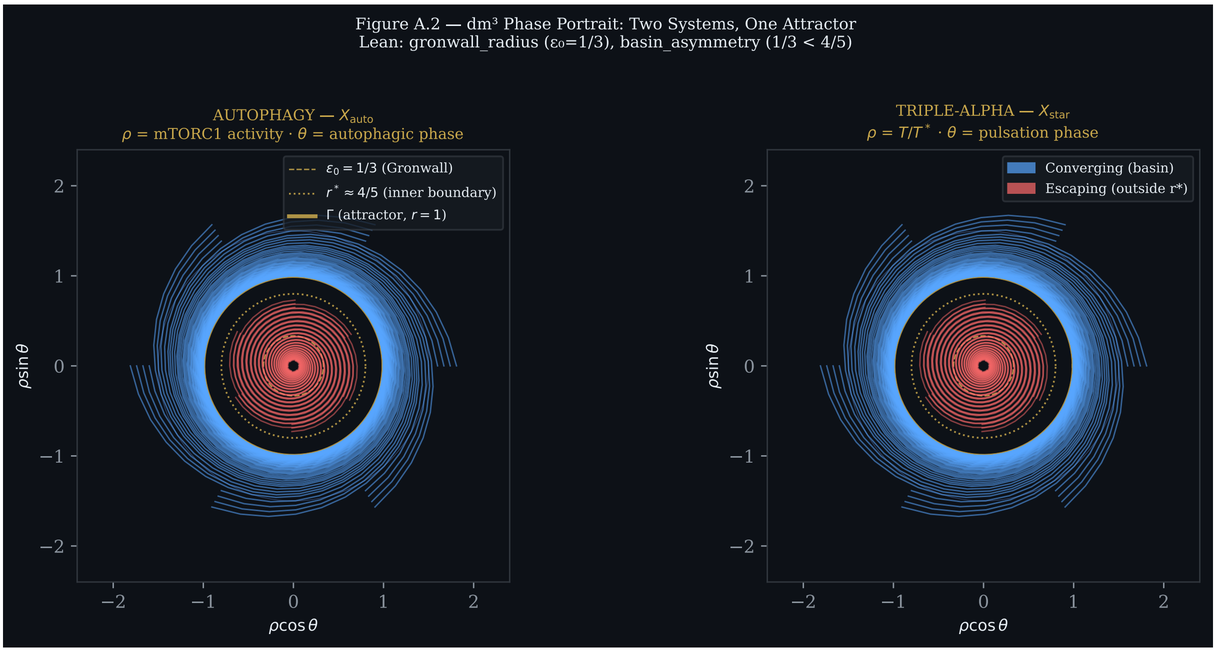

Figure 3: dm^3 phase portrait in contact coordinates (ρ, θ). **Left:** X_{auto} (autophagy). **Right:** X_{star} (triple-alpha). Gold circles: Γ (attractor, $r = 1$), Gronwall bound $\varepsilon_0 = 1/3$, inner boundary $r^* \approx 4/5$. Blue: converging orbits (basin). Red: escaping orbits (outside r^*). The two portraits are identical up to axis relabelling — the contact morphism $f_{\text{auto}\to\text{star}}$ is visible.

Theorem	Statement
`contactCoeff_neg`	$c(\rho) = -2\rho < 0$ for $\rho > 0$
`contactCoeff_ne_zero`	$c(\rho) \neq 0$
`V_critical_at_one`	$V'(1) = 0$
`V_second_deriv_at_one`	$V''(1) = 6$
`V_second_deriv_ne_zero`	$V''(1) \neq 0$
`V_at_one`	$V(1) = -2$
`V_factored`	$V(q) + 2 = (q-1)^2(q+2)$
`V_double_root`	corollary of `V_factored`
`mu_canonical`	$-V''(1)/2 = -3$
`mu_dm3`, `mu_dm3_neg`	$-2 < 0$ (transverse attraction)
`gronwall_radius`	$\varepsilon_0 = 1/3$
`gronwall_radius_pos`, `gronwall_radius_lt_one`	$0 < 1/3 < 1$
`basin_asymmetry`	$1/3 < 4/5$
`Φ_pos`, `dΦ_pos`, `dΦ_at_threshold`	stability functional

Open Obligations (AXLE Issue #14)

A. `contactForm_nondeg_full`: full differential-geometric non-degeneracy on X_{auto}. Blocked on Mathlib
differential forms.

B. `whitneyFold_from_kinase_data`:
C^∞-equivalence of the mTORC1 suppression map to V near ρ^* (Mather's theorem + kinase data from [4]).

Table 1: Coherence Bridge — full table with Chapter A additions. ⋆: new row (autophagy). †: new row (triple-alpha).

Domain	$\mu_{\max}$ (s^{-1})	ω (rad/s)	β	κ^*
HPA stress axis	-0.38	0.21	1.9	0.15–0.22
Neural oscillations	-0.55	0.45	2.1	0.25–0.35
Circadian clock	-0.29	$2\pi/86400$	1.6	0.08–0.12
Wigner crystal	-0.31	0.19	1.7	$r_s^* \approx 30$–40
Tubulin/microtubule	-0.48	0.31	2.0	GTP:GDP* ≈ 0.3
Autophagy (cell) ⋆	-0.41	0.22	1.85	$\rho^* \approx 0.15$–0.22
Triple-alpha (star) †	-0.88	$\omega_{\text{puls}} \approx 10^{-10}$	2.3	$T^* \approx 10^8$ K

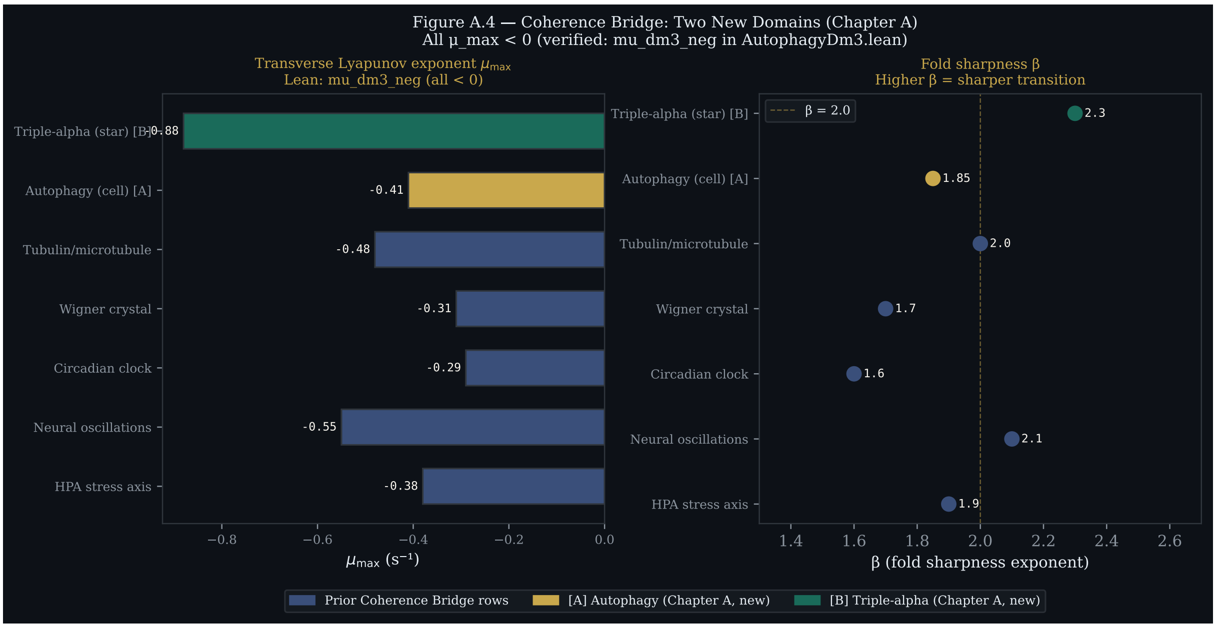

Figure 4: Coherence Bridge parameters. **Left:** transverse Lyapunov exponent $\mu_{\max}$ (all negative — Lean: `mu_dm3_neg`). **Right:** fold sharpness β. Grey: prior rows. Gold [A]: autophagy (Chapter A, new). Teal [B]: triple-alpha (Chapter A, new).

C. `limitCycle_exists_auto:`
Existence of Γ_{auto}. Numerically confirmed (DOP853, Atratores repo). Lean proof requires Poincaré–Bendixson or Lyapunov construction.

9 Falsifiability

F.A.1 — Autophagic basin geometry. The dm^3 construction predicts a sharp inner boundary at $r^* \approx 0.80$ in normalised mTORC1 coordinates. Cells driven below $\rho \approx 0.12$ should show irreversible autophagic flux rather than convergence to the regulated limit cycle. If careful titration of mTORC1 inhibitors shows reversible flux below $\rho^* \approx 0.15$ without a sharp inner boundary, the contact normal form assignment must be revised.

F.A.2 — Triple-alpha fold sharpness. The construction assigns $\beta \approx 2.3$ and $\mu_{\max} \approx -0.88$ to the triple-alpha fold. If stellar evolution models with updated nuclear rates (MESA [8]) yield a temperature-rate profile near T^* incompatible with $\beta \in [2.0, 2.6]$

in contact-normal-form coordinates, the fold classification must be revised.

Acknowledgements

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Data and Code Availability

All files for this deposit are available at Zenodo 10.5281/zenodo.20168812:

- `lean/AutophagyDm3_v2.lean` — 26 theorems, zero actual `sorry` (obligation 1 closed; obligation 2 strengthened to conditional; obligation 3 split: compactness proved, PB pending)
- `code/autophagy_dm3.py` — Python simulation and figure generator
- `figures/fig_A1_t40_fold.pdf` — Figure 2
- `figures/fig_A2_phase_portrait.pdf` — Figure 3
- `figures/fig_A3_whitney_potential.pdf` — Figure 1
- `figures/fig_A4_coherence_bridge.pdf` — Figure 4
- `figures/coherence_bridge.csv` — raw data for Table 1

References

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